

Selective Median Front Groups in Bottom-Disciplined B_N

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Selective Median Front Groups

This note summarizes autonomous discussion passes 36--38 of the APS/G2-ZOO ledger. The setting is the bottom-disciplined B_N family with APS orbit

$$T \mapsto a_1 \mapsto \cdots \mapsto a_{N+1} \mapsto s \mapsto s,$$

true bottom b , helper top U , and the order relations $b \leq x \leq U$ plus $s \leq a_{N+1}$.

Problem

The front $F_k = \{a_1, \dots, a_k\}$ can carry orthogonal idempotents in the front-shifted tensor, but same-order group fronts are rigid: the diagonal residual fiber strands the incomparable pair $\{T, e_G\}$, where T is the global monoid unit and e_G is the local group identity. With no additional join below U , the fiber has no principal maximum.

Capped Front

A full cap c above the entire front does not repair the obstruction. In B_N^{cap} , antitonicity forces

$$\boxtimes c = b,$$

so the APS orbit profile is preserved. But monotonicity ejects the cap from the relevant fibers: if $p \leq c$ and $p \otimes p = p$, then

$$p \otimes c \geq p \otimes p = p.$$

For any target t with $p \not\leq t$, the cap cannot lie in $\{x : p \otimes x \leq t\}$. Hence ceilings above the whole front cannot repair fibers whose obstruction sits at or below the front.

Selective Median

The successful repair is the single selective median

$$m = T \vee e_G, \quad b, T, e_G \leq m \leq U,$$

with m incomparable to non-identity front atoms and the tail. For abelian cyclic fronts $\mathbb{Z}/2, \dots, \mathbb{Z}/5$, the checker `code/scripts/check-selective-median-bound.py` verifies a commutative fully residuated tensor in which

$$a_j \setminus a_j = \{b, T, e_G, m\}$$

has principal maximum m .

Non-Abelian Test

Pass 38 drops commutativity and checks the smallest non-abelian front, $F_6 \cong S_3$. The tensor keeps the group product on the front, makes m a two-sided identity on front atoms, and sends front-tail, tail-front, median-tail, tail-median, and nonzero U interactions to U .

The report `artifacts/reports/noncommutative-selective-median-check.json` verifies associativity, two-sided monotonicity, unit, and principality of every left and right residual fiber. Both diagonal fibers are principal:

$$a_j \backslash a_j = \{b, T, e_G, m\} = a_j / a_j,$$

with maximum m . The forced antitone value remains $\boxtimes m = b$. The no-median and full-cap controls fail with the predicted residual and monotonicity obstructions.

Current Conjecture

Every finite group G should fit as the front of a two-residuated $B_N^{\text{med}}(G)$ after adjoining the single median $m = T \vee e_G$. The remaining proof task is to write the uniform finite-group theorem: associativity follows from the group block plus collapse blocks, while left and right residuals follow from bijectivity of left and right translations in G .